

ACCRA TECHNICAL UNIVERSITY

Department of Computer Science

BCS 404: Introduction to Data Science with Python

PROJECT WORK

Academic Year: 2025/2026 Second Semester

Lecturer: **Dr. Joseph Dadzie**

Project Title

Exploratory Data Analysis, Statistical Analysis and Machine Learning using a Real-World Dataset

Background

The purpose of this project is to enable students to demonstrate practical competence in applying Python libraries for data visualization, statistical analysis, and introductory machine learning on a real-world dataset.

Dataset

Download the following dataset from Kaggle:

Titanic Dataset

<https://www.kaggle.com/competitions/titanic/data>

Students should register for a free Kaggle account if necessary.

Software Requirements

Students must use

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn

Project Tasks

Task 1: Data Acquisition

1. Download the dataset from Kaggle.
2. Import the dataset into Python.
3. Display:
 - Dataset dimensions
 - Column names
 - First five observations
 - Data types

Task 2: Data Cleaning

Perform the following:

- Detect missing values.
- Handle missing values appropriately.
- Detect duplicated observations.
- Remove duplicates if any exist.
- Explain all preprocessing decisions.

Task 3: Data Visualisation

Produce and interpret the following visualisations.

1. Histogram of passenger ages
2. Bar chart showing passenger class distribution
3. Boxplot of age by passenger class
4. Scatter plot of Age versus Fare
5. Correlation Heatmap
6. Pairplot of selected numerical variables

Each figure should contain

- Appropriate title
- Axis labels
- Brief interpretation

Task 4: Statistical Analysis

Perform the following analyses.

1. Descriptive statistics
2. Frequency distribution
3. Correlation analysis
4. Identify the strongest positive correlation.
5. Identify the strongest negative correlation.
6. Discuss three important statistical findings.

Task 5: Machine Learning

Develop a simple classification model that predicts passenger survival.

Perform the following:

1. Select suitable predictor variables.
2. Split the dataset into training and testing sets.
3. Train a Logistic Regression classifier.
4. Predict the testing data.
5. Compute
 - Accuracy
 - Confusion Matrix
 - Classification Report
6. Discuss model performance.

Task 6: Discussion and Conclusion

Write approximately two pages discussing

- Major findings
- Statistical insights
- Machine learning results
- Limitations of the study
- Recommendations

Report Format

The report should contain

1. Cover Page
2. Table of Contents
3. Introduction
4. Dataset Description
5. Methodology
6. Results
7. Discussion
8. Conclusion
9. References
10. Appendix (Python Code)

Submission Requirements

Students shall submit the following:

1. One printed hardcopy of the report.
2. A softcopy uploaded to a GitHub repository.
3. The GitHub repository must contain:
 - Jupyter Notebook (.ipynb)
 - Python scripts (.py) if applicable
 - Dataset (where licensing permits)
 - Report in PDF format
 - README.md describing the project
4. Submit the printed report to the Class Representative.
5. Submit the GitHub repository URL through the course communication platform or as instructed by the lecturer.

Submission Deadline

Sunday, 19 July 2026 at 11:59 p.m.

Marking Scheme

Activity	Marks
Data Acquisition	5
Data Cleaning	10
Data Visualisation	20
Statistical Analysis	25
Machine Learning Model	30
Discussion and Conclusion	10
Total	100

Academic Integrity

Students are expected to produce original work.

Any form of plagiarism, copying of code without proper acknowledgement, or submission of another student's work shall attract penalties in accordance with the University's Examination Regulations.

Dr. Joseph Dadzie

Course Lecturer

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